In [1]:
```
from __future__ import division
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
#import healpy as hp
#import astropy.units as u
#import astropy.constants as c
#import scipy.signal as sig
#import scipy.special as spec
import math

%matplotlib inline
%config InlineBackend.figure_format = 'retina'
```

In [2]:
```
matplotlib.rcParams.update({'font.size':15})
matplotlib.rcParams['figure.dpi'] = 150
#matplotlib.rcParams['figure.figsize'] = [5,3]
#matplotlib.rcParams['figure.figsize'] = [2.5,2.5]matplotlib.rcParams['text.
matplotlib.rcParams['text.usetex'] = True
matplotlib.rcParams['axes.unicode_minus'] = False
```

In [3]:
```
# define some units
km = 1e3 # m
sec = 1 # s
G = 6.6743e-11 # MKS
yr = 365*24*60*60 # s
ns = 1e-9*sec
c = 299792458 # m/s
AU = 149597870700 #m
eV = 1.6022e-19 # J
MeV = 1e6*eV
hbar = 1.05457e-34 # J.s
kB = 1.38e-23 # J/K
lyr = c*yr
fref = 1/yr
pc = 3.26*lyr
kpc = 1000*pc
Mpc = 1e6 * pc
H0 = 70*km/sec/Mpc
Msol = 2e30 # kg
erg = 1e-7 # J
cm = 1e-2 # m
g = 1e-3 # kg
NA = 6.02e23 # Avogadros number

# E&M constants
q_p = 1.6022e-19 # charge of proton (coulomb)
m_p = 1.67e-27 # mass of proton (kg)
m_e = 9.11e-31 # mass of electron (kg)
epsilon_0 = 8.85e-12 # permitivity of free space (MKS = C^2/N m^2)
mu_0 = 4*np.pi*1e-7 # permeability of free space (MKS T m/A)
ke = 1/(4*np.pi*epsilon_0)
```

```
print('H0 =', H0)
```

```
H0 = 2.2711882363160234e-18
```

In [4]:
```
# planck mass, etc.
M_P = np.sqrt(hbar*c/G)
E_P = M_P * c**2 # = np.sqrt(hbar*c**5/G)
T_P = E_P/kB # = np.sqrt(hbar*c**5/(kB**2 G))
t_P = hbar/E_P # = np.sqrt(hbar*G/c**5)
L_P = c*t_P # = np.sqrt(hbar*G/c**3)

rho_P = E_P/L_P**3 # = c**7/(hbar*G**2)

print('M_P =', M_P, 'kg')
print('T_P =', T_P, 'Kelvin')
print('t_P =', t_P, 's')
print('L_P =', L_P, 'm')
print('E_P =', E_P, 'J or ', E_P/eV, 'eV')
print('rho_P =', rho_P, 'J/m^3 or', rho_P*cm**3/erg, 'erg/cm^3')
```

```
M_P = 2.1764324670802432e-08 kg
T_P = 1.4174492398255918e+32 Kelvin
t_P = 5.391241802170761e-44 s
L_P = 1.6162536315451223e-35 m
E_P = 1956079950.9593167 J or  1.2208712713514646e+28 eV
rho_P = 4.6329547731952086e+113 J/m^3 or 4.6329547731952093e+114 erg/cm^3
```

converting between units

In [5]:
```
# mass of electron in kg
m_e = 0.511 * MeV /c**2
print('m_e =', m_e, 'kg')
```

```
m_e = 9.10953527022454e-31 kg
```

problems (Chpt 23)

In [6]:
```
# prob 1
q = 41
r = 1.9e3

F = ke*q**2/r**2

print(F, 'newtons')
```

```
4187042.5737902573 newtons
```

In [7]:
```
# prob 2
q1 =  5.3e-9
q2 = 6e-9
q3 = -3e-9
r12 = 0.325
r13 = 0.1
```

```
Fx = -ke*abs(q1)*abs(q2)/r12**2
Fy = -ke*abs(q1)*abs(q3)/r13**2
F = np.sqrt(Fx**2 + Fy**2)
theta = math.atan2(Fy, Fx)

print(Fx, Fy, 'newtons')
print(F, 'newtons')
print(theta*180/np.pi + 360, 'degrees')
```

```
-2.70711847842598e-06 -1.4296969464187205e-05 newtons
1.4551007742288393e-05 newtons
259.278030940609 degrees
```

In [8]:
```
# prob 4
A = 4
B = 4
C = 8
theta = np.pi/4

Ex = A+B*np.cos(theta)/2
Ey = C+B*np.sin(theta)/2
E = np.sqrt(Ex**2 + Ey**2)
theta = math.atan2(Ey, Ex)

print(E)
print(theta*180/np.pi, 'degrees')
```

```
10.860070234439291
60.09633807948923 degrees
```

In [9]:
```
# prob 5
m = 2e-3
L = 24.1e-2
E = 1e3
g = 9.8
theta = 16.7*np.pi/180

q = m*g*np.tan(theta)/E
print(q, 'C')
```

```
5.880281805204364e-06 C
```

In [10]:
```
# prob 7
Q = -7.5e-6
L = 15e-2
R = L/np.pi
Lambda = Q/L
E = abs(Lambda)/(2*np.pi*epsilon_0*R)
print(E, 'N/C')
```

```
18832391.713747647 N/C
```

In [11]:
```
# prob 9
m = m_p
q = q_p
v = 3.8e5
E = 8.2e3
```

```
dx = 4.5e-2

dt = dx/v

a = q*E/m
dy = (1/2)*a*dt**2

vx = v
vy = a*dt

print(dt, 'sec')
print(dy, 'm')
print(vx, 'm/s')
print(vy, 'm/s')
```

```
1.1842105263157895e-07 sec
0.005516224683596795 m
380000.0 m/s
93162.90576741254 m/s
```

In [12]:

```
# prob 10
m = 5.06e-3
Q = -7.56e-6
theta = 24.9*np.pi/180
g = 9.8

E = m*g*np.sin(theta)/abs(Q)
print(E, 'N/C')
```

```
2761.6830573104694 N/C
```

In [13]:

```
# prob 11
q1 = 0.728e-6
q2 = -0.54e-6
d = 3.6e-2
r = 4.8e-2

k = ke*q1*abs(q2)/(r**2 * d)

print(k, 'N/m')
```

```
42.61740766643855 N/m
```

In [14]:

```
# prob 12
E = 660
dv = 1.4e6
m = m_p
q = q_p

a = q*E/m
dt = dv/a
dx = (1/2)*a*dt**2
K = (1/2)*m*dv**2

print(a, 'm/s^2')
print(dt, 'sec')
```

```
print(dx, 'm')
print(K, 'J')
```

```
63320479041.91617 m/s^2
2.2109750608065427e-05 sec
15.4768254256458 m
1.6366000000000002e-15 J
```

problems (Chpt 24)

In [15]:
```
# prob 2
q = 16.2e-6
R = 25.5e-2
Phi = q/epsilon_0
print(Phi, 'N m^2/C')
print(Phi/2, 'N m^2/C')
```

```
1830508.4745762711 N m^2/C
915254.2372881356 N m^2/C
```

In [16]:
```
# prob 3
Lambda = -90.2e-6

ra = 10e-2
Ea = abs(Lambda)/(2*np.pi*epsilon_0 * ra)

rb = 23.5e-2
Eb = abs(Lambda)/(2*np.pi*epsilon_0 * rb)

rc = 150e-2
Ec = abs(Lambda)/(2*np.pi*epsilon_0 * rc)

print(Ea, 'N/C')
print(Eb, 'N/C')
print(Ec, 'N/C')
```

```
16221215.66880108 N/C
6902644.965447269 N/C
1081414.377920072 N/C
```

In [17]:
```
# prob 4
q = -0.654e-6
m = 18.6e-3
g = 9.8

sigma = 2*epsilon_0 * m* g/q
print(sigma, 'C/m^2')
```

```
-4.933266055045872e-06 C/m^2
```

In [18]:
```
# prob 5
Q = 34e-6
r = 25e-2
```

```
E = ke*Q/r**2
print(E, 'N/C')
```

```
4891541.753784806 N/C
```

In [19]:
```
# prob 6
ri = 20e-2
ro = 32e-2
q = -60e-9
rho = -2.92e-6

Q = rho*(4*np.pi/3)*(ro**3-ri**3)
E = ke*abs(Q + q)/ro**2
v = np.sqrt(abs(q_p)*E*ro/m_p)

print(Q, 'C')
print(E, 'N/C')
print(v, 'm/s')
```

```
-3.029440309130761e-07 C
31870.33047841285 N/C
989164.2246510898 m/s
```

In [20]:
```
# prob 7
R = 4.9e-2
Lambda = 31.2e-9
rb = 20e-2
rc = 200e-2

Eb = Lambda/(2*np.pi*epsilon_0*rb)
Ec = Lambda/(2*np.pi*epsilon_0*rc)

print(Eb, 'N/C')
print(Ec, 'N/C')
```

```
2805.4430646706974 N/C
280.54430646706976 N/C
```

In [21]:
```
# prob 8
E = 82e3
s = 55e-2
A = s**2

sigma_minus = -E*epsilon_0
sigma_plus = -sigma_minus

Q_minus = sigma_minus*A
Q_plus = -Q_minus

print(sigma_plus, 'C/m^2')
print(Q_plus, 'C')
```

```
7.257e-07 C/m^2
2.1952425000000003e-07 C
```

problems (Chpt 25)

In [22]:
```
# prob 1
dV = 600
ds = 5.4e-3
d = 2.92e-3
q = q_p

dx = ds-d

E = dV/ds
F = q*E
W = F*dx

print(E, 'N/C')
print(F, 'N')
print(W, 'J')
```

```
111111.11111111111 N/C
1.7802222222222222e-14 N
4.414951111111112e-17 J
```

In [23]:
```
# prob 2
v = 2.75e7
m = m_e
q = -q_p

dV = -(1/2)*m_e*v**2/q
print(dV, 'volts')
```

```
2149.8833004953526 volts
```

In [24]:
```
# prob 3
r1 = 0.1e-2
r2 = 0.87e-2

q = -q_p

V1 = ke*q/r1
V2 = ke*q/r2
dV = V2-V1

print(V1, 'volts')
print(dV, 'volts')
```

```
-1.4406669481459588e-06 volts
1.2750730460602163e-06 volts
```

In [25]:
```
# prob 4
d = 3.5e-2
q1 = -19e-9
q2 = 28.5e-9
VA = (ke/d)*(q1+q2)
VB = (ke/(d/2))*(q1+q2)
```

```
print(VA, 'volts')
print(VB, 'volts')
```

```
2440.6327027812836 volts
4881.265405562567 volts
```

In [26]:

```
# prob 5
Q = 90e-9
y = 12e-2
x = 9e-2

q1 = Q
q2 = -Q
q3 = Q/2
q4 = 2*Q

r12 = 2*y
r13 = y
r23 = y
r34 = x
r14 = np.sqrt(x**2 + y**2)
r24 = r14

m = 2.34e-13

#U3 = -ke*Q**2/(2*y)
#U4 = U3 + ke*Q**2/x
U3 = ke*(q1*q2/r12 + q1*q3/r13 + q2*q3/r23)
U4 = U3 + ke*(q1*q4/r14 + q2*q4/r24 + q3*q4/r34)

K = U4-U3
v = np.sqrt(2*K/m)
print(U3, 'J')
print(U4, 'J')
print(K, 'J')
print(v, 'm/s')
```

```
-0.000303473408437936 J
0.0005057890140632268 J
0.0008092624225011628 J
83167.13799266078 m/s
```

In [27]:

```
# prob 6
m1=0.4
r1=0.3e-2
q1=-2e-6
m2=0.7
r2=0.5e-2
q2=5.5e-6
d=1
v1 = np.sqrt(-2*(m2/m1)*(1/(m1+m2))*ke*q1*q2*(d-r1-r2)/(d*(r1+r2)))
v2 = (m1/m2)*v1
print(v1, v2, 'm/s')
```

```
6.246953847592036 3.5696879129097354 m/s
```

In [28]:
```
# prob 9
Q=2.1e-6
q=1.2e-18
x=0.8
V=2*ke*Q/x
print(V, 'volts')
```

```
47206.97464590115 volts
```

problems (Chpt 26)

In [29]:
```
# prob 1
dV = 12
Q = 12e-6

C = Q/dV
print(C, 'F')

Q = 144e-6
dV = Q/C

print(dV, 'volts')
```

```
1e-06 F
144.0 volts
```

In [30]:
```
# prob 2
a=7.1e-2
b=13.2e-2
C = 4*np.pi*epsilon_0*a*b/(b-a)
print(C, 'F')
Q=4e-6
dV=Q/C
print(dV, 'volts')
```

```
1.7086597127381985e-11 F
234101.61603154047 volts
```

In [31]:
```
# prob 3
L=45
a=2.58e-3/2
b=7.27e-3/2
Q=8.1e-6
C = 2*np.pi*epsilon_0*L/np.log(b/a)
print(C, 'F')
dV=Q/C
print(dV, 'volts')
```

```
2.4154039732635392e-09 F
3353.476308584439 volts
```

In [32]:
```
# prob 4
A=2.10e-4
d=1e-3
dV=9
```

```
C = epsilon_0*A/d
print(C, 'F')
Q = C*dV
print(Q, 'C')
E = dV/d
print(E, 'V/m')
```

```
1.8585000000000002e-12 F
1.6726500000000002e-11 C
9000.0 V/m
```

In [33]:

```
# prob 6
C1=3e-6
C2=16e-6
dV=9
Ceq = 1/(1/C1+1/C2)
print(Ceq, 'F')
Q = Ceq*dV
print(Q, 'C')
dV1 = Q/C1
dV2 = Q/C2
print(dV1, dV2, 'volts')
```

```
2.5263157894736844e-06 F
2.273684210526316e-05 C
7.578947368421053 1.4210526315789476 volts
```

In [34]:

```
# prob 8
C1=6e-6
C2=13e-6
C3=3e-6
Cs=C1*C2/(C1+C2)
print(Cs, 'F')
Ceq = 1/(1/(C3+2*Cs) + 1/(2*C2))
print(Ceq, 'F')
Ctop = C3+2*Cs
Cbot = 2*C2
dV=60
dVtop = (Ceq/Ctop)*dV
dVbot = (Ceq/Cbot)*dV
print(dVtop, dVbot, 'volts')
Q3 = C3*dVtop
print(Q3, 'C')
```

```
4.1052631578947365e-06 F
7.833097595473833e-06 F
41.923620933521924 18.07637906647808 volts
0.00012577086280056579 C
```

In [35]:

```
# prob 9
C1=8e-6
C2=9e-6
dV=22
Q1i = C1*dV
print(Q1i, 'C')
Ceq = C1+C2
dVf = Q1i/Ceq
```

```
print(dVf, 'volts')
Q1f = C1*dVf
Q2f = C2*dVf
print(Q1f, Q2f, 'C')
```

```
0.000176 C
10.352941176470589 volts
8.28235294117647e-05 9.317647058823531e-05 C
```

In [36]:
```
# prob 10
C=15e-6
dV=45.5
Q = C*dV
Ui=2*(1/2)*C*dV**2
print(Ui, 'J')
dVf = (4/3)*(Q/C)
print(dVf, 'volts')
Uf = (4/3)*Ui
print(Uf, 'J')
```

```
0.03105375 J
60.666666666666664 volts
0.041405 J
```

In [37]:
```
# prob 11
kappa = 2
A=1.7e-4
d=0.08e-3
C = kappa*epsilon_0*A/d
print(C, 'F')
E_max = 30e3/1e-3
dV_max = E_max * d
print(dV_max, 'volts')
```

```
3.76125e-11 F
2400.0 volts
```

In [38]:
```
# additional problems (current, resistance)
```

In [39]:
```
# thermal velocity of electrons at room temperature
# (1/2) m_e v_th^2 = (3/2) kB T

T = 273 + 20 # Kelvin
v_th = np.sqrt(3*kB*T/m_e)
print(v_th, 'm/s')
```

```
115394.70198480072 m/s
```

In [40]:
```
# drift velocity
A = 3.3e-6 # m^2
I = 10 # amps
q = q_p
Mm = 63.5 # g/mole
rho = 8.96 # g/cm^3
n = rho * (1/Mm) * NA * 1e6
print(n, 'electrons / m^3')
```

```
v_d = I/(A*q*n)
print(v_d, 'm/s')
```

```
8.49436220472441e+28 electrons / m^3
0.00022265812988681168 m/s
```

problems (Chpt 27)

In [41]:
```
# prob 1
r=1.5e-3
I=3.6
n=8.46e28
q=q_p
A=np.pi * r**2
v_d = I/(A*q*n)
print(v_d, 'm/s')
```

```
3.757361832961837e-05 m/s
```

In [42]:
```
# prob 2
I=5.6
r1=0.26e-2
A1 = np.pi*r1**2

J1 = I/A1
print('J1=', J1, 'A/m^2')

A2 = 3*A1
r2 = np.sqrt(A2/np.pi)
print('r2=', r2, 'm')

J2 = I/A2
print('J2=', J2, 'A/m^2')
```

```
J1= 263688.6631108325 A/m^2
r2= 0.004503332099679081 m
J2= 87896.22103694417 A/m^2
```

In [43]:
```
# prob 3
L = 50
V = 9.11
d = 2e-3
r = d/2
A = np.pi*r**2
I = 2.60
R = V/I

# R = rho*L/A
rho = R*A/L
print(rho, 'ohms.m')
```

```
2.201531467246385e-07 ohms.m
```

In [44]:
```
# prob 4
rho_tungsten = 5.25e-8
rho_gold = 2.35e-8
```

```
ratio = np.sqrt(rho_tungsten/rho_gold)
print(ratio)
```

```
1.494671386356041
```

In [45]:

```
# prob 5
alpha = 4.5e-3 # 1/K
R0 = 20.9
R = 147
T0 = 20 # celsius
T = T0 + (R/R0 - 1)/alpha
print(T, 'celsius')
```

```
1360.7761828814462 celsius
```

In [46]:

```
# prob 6
R = 14
r = 1.1e-3
rho_C = 3.5e-5
rho_N = 150e-8
alpha_C = -0.5e-3
alpha_N = 0.4e-3
A = np.pi*r**2

l2 = R*A/(rho_N*(1-alpha_N/alpha_C))
l1 = -l2*(rho_N/rho_C)*(alpha_N/alpha_C)

print(l1, 'm')
print(l2, 'm')
```

```
0.6757914863722045 m
19.710585019189296 m
```

In [47]:

```
# prob 7
hp = 745.7 # watt
P_mech = 1500*hp
P_elec = 0.8*P_mech
V = 2020
I = P_elec/V

print(I, 'amps')
```

```
442.990099009901 amps
```

In [48]:

```
# prob 8
P = 1.04e3
V = 120
I = P/V
R = P/I**2

print(I, 'amp')
print(R, 'ohm')
```

```
8.666666666666666 amp
13.846153846153848 ohm
```

In [49]:
```
# prob 9
c_w = 4184 # J/(kg . C)
V = 240
m = 131
dT = 31
dt = 27*60
R = V**2 * dt/(m*c_w*dT)
print(R, 'ohm')
```

```
5.491776225185425 ohm
```

In [50]:
```
# prob 10
dP = (40-11)/1000 # kW
cost_rate = 0.114 # dollars/kWh
dt = 130 # hr
savings = cost_rate*dP*dt

print(savings, 'dollars')
```

```
0.42978000000000005 dollars
```

problems (Chpt 28)

In [51]:
```
# prob 1
E = 12.6
ri = 0.056
R = 4.6
V1 = E * (R/(ri+R))
print(V1, 'volts')
I = 35
V2 = I*R
print(V2, 'volts')
```

```
12.448453608247423 volts
161.0 volts
```

In [52]:
```
# prob 3
V = 25
R1 = 10
R2 = 10
R3 = 5
R4 = 5
R5 = 18
R45 = R4+R5
Rp = 1/(1/R2 + 1/R3 + 1/R45)
print(Rp, 'ohms')
I = V/(R1 + Rp)
Vp = I*Rp
print(Vp, 'volts')
I5 = Vp/R45
print(I5, 'amps')
```

```
2.911392405063291 ohms
5.637254901960785 volts
0.2450980392156863 amps
```

In [53]:
```
# prob 4
i0 = 1.09e-3
ia = 1.22e-3
ib = 2.01e-3
E = 3
C1 = E/i0
C2 = E/ia
C3 = E/ib
print(C1, C2, C3, 'ohms')
R3 = C1-C3
R2 = 2*(C1-C2)
R1 = C3-R2
print(R1, R2, R3, 'ohms')
```

```
2752.293577981651 2459.016393442623 1492.5373134328358 ohms
905.9829443547799 586.554369078056 1259.7562645488154 ohms
```

In [54]:
```
# prob 5
E1 = 40
E2 = 360
E3 = 80
R1 = 80
R2 = 20
R3 = 70
R4 = 200
M = np.array([[1, 1, 1, 1], [R1, 0, 0, -R4], [-R1, R2, 0, 0], [0, -R2, R3, 0
y = np.array([0, E1, -E1-E2, E2+E3])
Minv = np.linalg.inv(M)
x = Minv@y
print(x, 'amps')
V3 = x[2]*R3
print(V3, 'volts')
```

```
[ 3. -8.  4.  1.] amps
280.0 volts
```

In [55]:
```
# prob 6
R = 3e6
C = 6e-6
E = 34
t = 10

tau = R*C
Qmax = C*E
i10 = (E/R)*np.exp(-t/tau)

print(tau, 'sec')
print(Qmax, 'coul')
print(i10, 'amps')
print(E/R)
```

```
18.0 sec
0.000204 coul
6.502538768357572e-06 amps
1.1333333333333334e-05
```

In [56]:
```
# prob 7
P1 = 2000
P2 = 500
P3 = 1000
V =120

I1 = P1/V
I2 = P2/V
I3 = P3/V

I_tot = I1 + I2 + I3
print(I1, I2, I3, 'amps')

print(I_tot, 'amps')
```

```
16.666666666666668 4.166666666666667 8.333333333333334 amps
29.16666666666667 amps
```

In [57]:
```
# prob 8
C1 = 3e-6
C2 = 2e-6
R = 500
V = 24

Ceq = C1+C2
Qeq = Ceq * V
tau = Ceq*R
print(tau, 'sec')

t = 3e-3
q3 = Qeq*np.exp(-t/tau)
V3 = q3/Ceq
Q1_3 = C1*V3
Q2_3 = C2*V3
i_3 = V3/R

print(Qeq/1e-6, 'microcoul')
print(q3/1e-6, 'microcoul')
print(V3, 'volts')
print(Q1_3/1e-6, 'microcoul')
print(Q2_3/1e-6, 'microcoul')
print(i_3/1e-3, 'mA')
```

```
0.0024999999999999996 sec
120.0 microcoul
36.14330542946424 microcoul
7.228661085892849 volts
21.68598325767855 microcoul
14.457322171785698 microcoul
14.457322171785698 mA
```

In [58]:
```
# prob 9
P = 45
V = 120
R = V**2/P
```

```
print(R, 'ohms')

Req = 3*R/2
I = V/Req
Ptot = V*I

print(I, 'amp')
print(Ptot, 'watts')

V1 = I*R
V2 = I*R/2
V3 = V2
print(V1, V2, V3, 'volts')
```

```
320.0 ohms
0.25 amp
30.0 watts
80.0 40.0 40.0 volts
```

problems (Chpt 29)

In [59]:

```
# prob 2
m = m_p
q = q_p
F = q*np.sqrt(4**2 + 5**2 + 14**2)
print(F, 'newtons')
```

```
2.4665554788453393e-18 newtons
```

In [60]:

```
# prob 3
m = m_p
q = q_p
v = 4.9e6
B = 0.24
theta = 57*np.pi/180

F = q*v*B*np.sin(theta)
a = F/m

print(F, 'newtons')
print(a, 'm/s^2')
```

```
1.5802123491394982e-13 newtons
94623493960448.98 m/s^2
```

In [61]:

```
# prob 4
m = m_e
q = -q_p
B = 2.17e-3
v = 1.35e7

R = m*v/(np.abs(q)*B)
T = 2*np.pi*R/v
```

```
print(R, 'm')
print(T, 's')
```

```
0.035371504201317454 m
1.6462645591893246e-08 s
```

In [62]:
```
# prob 6
m = m_p
q = q_p
R = 1.6
B = 0.33

omega = q*B/m
v_max = q*B*R/m

print(omega, 'rad/s')
print(v_max, 'm/s')
```

```
31660239.520958085 rad/s
50656383.23353294 m/s
```

In [63]:
```
# prob 7
m = 1.8e-26
q = q_p
E = 2.9e3
B = 0.045

R = m*E/(q*B*B)
print(R, 'm')
```

```
0.1608898875157769 m
```

In [64]:
```
# prob 8
m_by_L = 0.052 # kg/m
I = 1.6
g = 9.8

B = m_by_L*g/I
print(B, 'tesla')
```

```
0.3185 tesla
```

In [65]:
```
# prob 9
l = 14.4e-2
m = 15.8e-3
I = 4.75
g = 9.8

B = m*g/(I*l)
print(B, 'tesla')
```

```
0.22637426900584803 tesla
```

In [66]:
```
# prob 10
I = 12 # mA
C = 3.2
B = 0.67
```

```
R = C/(2*np.pi)
A = np.pi * R**2

mu = I*A
tau = mu*B

print(mu, 'mA m^2')
print(tau, 'mN m')
```

```
9.778479703566049 mA m^2
6.551581401389253 mN m
```

In [67]:

```
# prob 11
m = m_e
q = q_p
theta = 85*np.pi/180
B = 0.139
v = 4.75e6

v_x = v*np.cos(theta)
v_perp = v*np.sin(theta)

T = 2*np.pi*m/(q*B)
p = v_x*T
r = m*v_perp/(q*B)

print(p, 'm')
print(r, 'm')
```

```
0.00010639817538529626 m
0.00019355416880394605 m
```

In [68]:

```
4*np.pi*epsilon_0*.1
```

Out[68]:

```
1.112123799370787e-11
```

problems (Chpt 30)

In [69]:

```
# prob 1
l = 0.46
I = 10.6

a = l/2
R = 2*l/np.pi

# square
B1 = (2*np.sqrt(2)/np.pi) * mu_0 * I/l
print(B1, 'T')

# circle
B2 = (np.pi/4) * mu_0 * I/l
print(B2, 'T')

print(B1/B2)
```

```
2.6070719584617058e-05 T
2.2743001445988517e-05 T
1.1463183365015128
```

In [70]:
```
# prob 2
R = 12e-2
I = 2.9

B1 = mu_0*I/(2*R)
B2 = mu_0*I/(2*np.pi*R)

B = B1+B2
print(B, 'T')
```

```
2.0017697825684e-05 T
```

In [71]:
```
# prob 4
R = 0.6
I = 5
theta = 30*np.pi/180

B = mu_0*I/(24*R)
print(B, 'T')
```

```
4.3633231299858245e-07 T
```

In [72]:
```
# prob 5
d = 10.7e-2
I = 5.35

B0 = 2*mu_0*I/(np.pi * d)
print(B0, 'T')

B1 = mu_0*I/(4 * np.pi * d)
print(B1, 'T')

B2 = mu_0*I/(12 * np.pi * d)
print(B2, 'T')
```

```
3.9999999999999996e-05 T
4.9999999999999996e-06 T
1.6666666666666667e-06 T
```

In [73]:
```
# prob 6
d = 3.55e-2
F_by_L = 1.8e-4
I1 = 4.75

I2 = 2*np.pi*d*F_by_L/(mu_0 * I1)
print(I2, 'A')
```

```
6.726315789473684 A
```

In [74]:
```
# prob 7
I1 = 9
I2 = 10
d = 0.1
a = 0.15
```

```
l = 0.5

F = ((mu_0 * I1 * I2 * l)/(2*np.pi)) * (a/(d*(d+a)))
print(F, 'N')
```

5.399999999999999e-05 N

In [75]:
```
# prob 8
I1 = 1.2
I2 = 3.16
d = 1e-3

Ba = mu_0*I1/(2*np.pi*d)
print(Ba, 'T')

Bb = mu_0*(I1-I2)/(2*np.pi*3*d)
print(Bb, 'T')
```

0.00024 T
-0.00013066666666666668 T

In [76]:
```
# prob 9
ri = 0.7
ro = 1.3
N = 840
I = 12e3

Bi = mu_0*I*N/(2*np.pi*ri)
print(Bi, 'T')

Bo = mu_0*I*N/(2*np.pi*ro)
print(Bo, 'T')
```

2.88 T
1.5507692307692307 T

In [77]:
```
# prob 10
d = 0.1e-2
D = 10e-2
L = 66.1e-2
B = 7.7e-3
rho = 1.68e-8 # resistivity copper

r = d/2
R = D/2
n = 1/d

I = B/(mu_0*n)
print(I, 'A')

R = 4*rho*L*D/d**3
print(R, 'ohms')

P = I**2 * R
print(P, 'W')
```

```
6.127465309037971 A
4.4419200000000005 ohms
166.77557813951714 W
```

problems Chpt 31

In [78]:
```
# prob 1
A = 9*(1e-2)**2
R = 2.90
dB = 2.10-0.5
dt = 1.06

dBdt = dB/dt
I = A*dBdt/R

print(dBdt, 'T/s')
print(I, 'A')
```

```
1.509433962264151 T/s
0.00046844502277163316 A
```

In [79]:
```
# prob 2
N = 24
d = 0.93
A = np.pi*(d/2)**2
B = 44e-6
dt = 0.2

emf = N*2*B*A/dt

print(emf, 'volts')
```

```
0.007173311603277106 volts
```

In [80]:
```
# prob 3
r1 = 5e-2
R = 4.25e-4
n = 1010
r2 = 3e-2
dIdt = 270

A = np.pi*r2**2
I_ind = 0.5*A*mu_0*n*dIdt/R
B_ind = mu_0*I_ind/(2*r1)

print(I_ind, 'A')
print(B_ind, 'T')
```

```
1.1399044744279347 A
1.4324462090627735e-05 T
```

In [81]:
```
# prob 4
b = 20
h = 1e-2
w = 17e-2
```

```
L = 1.5

emf = (mu_0/(2*np.pi))*b*L*np.log((h+w)/h)
print(emf, 'volts')
```

1.734223054737699e-05 volts

In [82]:
```
# prob 6
m = 0.26
l = 1.2
theta = 33*np.pi/180
R = 3.3
B = 0.5

v = m*g*R*(np.sin(theta)/(np.cos(theta)**2))/(B**2 * l**2)
print(v, 'm/s')
```

18.08576247319027 m/s

In [83]:
```
# prob 7
B = 0.33
R = 9.2
l = 0.32
I_ind = 8.7e-3

v = I_ind*R/(B*l)
print(v, 'm/s')

P = I_ind**2 * R
print(P, 'watts')
```

0.7579545454545452 m/s
0.0006963479999999998 watts

In [84]:
```
# prob 8
r2 = 0.02
t = 4.9
dBdt = 2*0.053*t

E = -0.5*r2*dBdt
print(E, 'N/C')
```

-0.005194 N/C

In [85]:
```
# prob 9
f = 65
s = 15e-2
B = 0.8
A = s**2
R = 2

omega = 2*np.pi * f
Phi = B*A

print(omega, 'rad/s')
print(Phi, 'T m^2')

emf = -Phi*omega
```

```
print(emf, 'volts')

I_ind = emf/R
print(I_ind, 'A')

P = I_ind**2 * R
print(P, 'W')

tau = I_ind*A*B
print(tau, 'N.m')
```

```
408.4070449666731 rad/s
0.018 T m^2
-7.351326809400115 volts
-3.6756634047000576 A
27.021002929302437 W
-0.06616194128460104 N.m
```

Problems Chpt 32

In [86]:
```
# prob 1
N = 440
l = 25e-2
r = 1.7e-2

n = N/l
A = np.pi*r**2

L = mu_0 * n**2 * A * l
print(L, 'H')

emf = 70e-3
didt = emf/L
print(didt, 'A/s')
```

```
0.0008835333025323362 H
79.22734751408886 A/s
```

In [87]:
```
# prob 2
I = 36.5e-3
N = 400
l = 11.5e-2
d = 16.5e-3

A = np.pi*(d/2)**2
n = N/l

B = mu_0 * I * n
print(B, 'T')

Phi_tot = N * B * A
Phi_one = B*A
print(Phi_tot, 'T m^2')
print(Phi_one, 'T m^2')
```

```
L = mu_0 * n**2 * A * l
print(L, 'H')
```

```
0.00015953827040838598 T
1.3645285931711312e-05 T m^2
3.411321482927829e-08 T m^2
0.00037384345018387163 H
```

In [88]:
```
# prob 3
L = 70e-3

t=1
didt = 6*t-7
emf = L*didt
print(emf, 'V')

t=4
didt = 6*t-7
emf = L*didt
print(emf, 'V')
```

```
-0.07 V
1.1900000000000002 V
```

In [89]:
```
# prob 4
L = 4.65e-3
R2 = 445
tau = 14.9e-6
V = 24

Reff = L/tau
print(Reff, 'ohms')

R1 = R2*Reff/(R2-Reff)
print(R1, 'ohms')

I = V/Reff
print(I, 'A')
```

```
312.08053691275165 ohms
1044.8119161827817 ohms
0.07690322580645162 A
```

In [90]:
```
# prob 5
L = 140e-3
R = 5.2
emf = 6

tau = L/R
print(tau, 's')

i = 220e-3
t = tau*np.log(emf/(emf-R*i))
print(t, 's')

t = 10
i = (emf/R)*(1-np.exp(-t/tau))
```

```
print(i, 'A')

i = 160e-3
t = tau*np.log(emf/(R*i))
print(t, 's')
```

```
0.026923076923076925 s
0.00569542657036925 s
1.1538461538461537 A
0.05319144673739571 s
```

In [91]:

```
# prob 6
E = 147
B = 4.5e-5

uE = 0.5*epsilon_0*E**2
print(uE, 'J/m^3')

uB = 0.5*(1/mu_0)*B**2
print(uB, 'J/m^3')
```

```
9.5619825e-08 J/m^3
0.0008057218994027203 J/m^3
```

In [92]:

```
# prob 7
B = 4.9
d = 6.2e-2
l = 26e-2
A = np.pi*(d/2)**2

uB = 0.5*(1/mu_0)*B**2
print(uB, 'J/m^3')

UB = uB*A*l
print(UB, 'J')
```

```
9553275.459091019 J/m^3
7498.923250000001 J
```

In [93]:

```
# prob 8
M = 130e-6
i0 = 15
omega = 1.25e3

emf = M*i0*omega
print(emf, 'V')
```

```
2.4375 V
```

In [94]:

```
# prob 9
h = 0.4e-3
w = 1.1e-3
l = 2.7e-3

M = (mu_0/(2*np.pi)) * l * np.log((h+w)/h)
print(M, 'H')
```

```
7.137481535904527e-10 H
```

In [95]:
```
# prob 10
R = 10
L = 0.1
C = 1.1e-6
emf = 12

omega = 1/np.sqrt(L*C)
f = omega/(2*np.pi)
print(f, 'Hz')

Qmax = C*emf
print(Qmax, 'C')

Imax = omega*Qmax
print(Imax, 'A')

E = 0.5*C*emf**2
print(E, 'J')
```

```
479.87020887834814 Hz
1.32e-05 C
0.0397994974842648 A
7.92e-05 J
```

In [96]:
```
# prob 11
L = 3.3
C = 838e-12
Qmax = 128e-6
t = 4e-3
omega = 1/np.sqrt(L*C)

UC = 0.5*(Qmax**2/C)*np.cos(omega*t)**2
print(UC, 'J')

E = 0.5*(Qmax**2/C)
print(E, 'J')

UL = E-UC
print(UL, 'J')
```

```
6.043005835108511 J
9.77565632458234 J
3.732650489473828 J
```

Problems (Chpt 35)

In [97]:
```
# prob 1
c_meas = 2.995e8
d = 11.5e3
N = 720

omega = np.pi * c_meas/(N*d)
print(omega, 'rad/s')
```

```
113.63611108093515 rad/s
```

In [98]:

```
# prob 2
R = 1.5e8*1e3
dt = 22*60
c_meas = 2*R/dt

print(c_meas, 'rad/s')
```

```
227272727.27272728 rad/s
```

In [99]:

```
# prob 3
l = 632.8e-9
n = 1.64

f = c/l
print(f/1e14, 'e14 Hz')

c_glass = c/n
l_glass = c_glass/f

print(l_glass/1e-9, 'nm')
print(c_glass, 'm/s')
```

```
4.737554646017699 e14 Hz
385.8536585365853 nm
182800279.2682927 m/s
```

In [100…

```
# prob 4
n1 = 1
n2 = 1.48
n3 = 1.33
phi = 25.6*np.pi/180

theta = np.arcsin((n2/n1)*np.sin(phi))
print(theta*180/np.pi, 'degrees')

theta_p = np.arcsin((n2/n3)*np.sin(phi))
print(theta_p*180/np.pi, 'degrees')
```

```
39.75357017105567 degrees
28.738789826279657 degrees
```

In [101…

```
# prob 5
n1 = 1
n2 = 1.5
L = 1.27e-2
theta = 33*np.pi/180

phi = np.arcsin((n1/n2)*np.sin(theta))
print(phi*180/np.pi, 'degrees')

d = L*np.sin(theta-phi)/np.cos(phi)
print(d, 'm')

dt = (L/np.cos(phi))/c
print(dt, 's')
```

```
21.290250634674873 degrees
0.0027663067999197967 m
4.546552464143133e-11 s
```

In [102…

```
# prob 6
d = 5.51e-2
n = 1.33

h = d*np.sqrt((n**2-1)/(4-n**2))
print(h, 'm')
```

```
0.032346478830816414 m
```

In [103…

```
# prob 7
d = 6.81e-6
n = 1.34

theta = np.arcsin(np.sqrt(n**2-1))
print(theta*180/np.pi, 'degrees')
```

```
63.121105079429164 degrees
```

In [104…

```
# prob 8
theta1 = 45*np.pi/180
theta2 = 25.5*np.pi/180

n = np.sin(theta1)/np.sin(theta2)
print(n)

theta = np.arcsin(1/n)
print(theta*180/np.pi, 'degrees')
```

```
1.6424821251504682
37.50528687864152 degrees
```

In [105…

```
# prob 9
n1 = 1.6
n2 = 1.4
n3 = 1.2
n4 = 1.2
theta1 = 34*np.pi/180

theta2 = np.arcsin((n1/n4)*np.sin(theta1))
print(theta2*180/np.pi, 'degrees')

theta1 = np.arcsin(n4/n1)
print(theta1*180/np.pi, 'degrees')
```

```
48.20984676819403 degrees
48.59037789072913 degrees
```